

# Requirement elicitation model for global software development vendors

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## Abstract

Requirements elicitation is critical phase of software development that involve direct communication of clients and vendors. In global software development (GSD), importance of requirements elicitation becomes more critical due to geographical distance. The aim of this research is to design and implement requirement elicitation model (REM) where GSD organization can evaluate their organizations and can classify their level of implementation where organization lies. Systematic literature review (SLR) was conducted for the identification of critical success factors (CSFs) from a sample of 70 research articles. Furthermore, the identified CSFs are organized into five levels based on Capability Maturity Model Integration (CMMI) and Software Outsourcing Vendors' Readiness Model (SOVRM). A second SLR was conducted for identifying all possible practices for the implementation of CSFs. To validate the findings of SLR and to find more practices in the view of industrial practitioners for the identified CSFs, a questionnaire survey was conducted in the GSD organizations in which 35 experts, from different countries, contributed. For REM evaluation, we have conducted two case studies. The outcome analysis after conducting case studies confirms that our assessment framework is easier to understand, apply, and successfully critic the strengths and weaknesses of organizations during requirements elicitation in GSD.

## KEYWORDS

case study, critical success factors, empirical study, requirements elicitation, systematic literature review

## 1 | INTRODUCTION

Global software development (GSD) faces a lot of challenges during requirements elicitation phase of software development. Success factors as identified by different researchers play significant role in reducing the effect of these challenges. According to Aranda et al.,<sup>1</sup> effective communication and proper channels for communication among different parties in GSD are necessary as stakeholders involve in GSD have geographical difference that makes it difficult to communicate properly specially when there exist challenges such as time zone difference, culture variation, and language barrier. Even people living in one country can have different languages and culture norms. For communication, we need strong communication channels and technologies.<sup>2</sup> Today in modern era, implementation of collaborative tools and technologies play a big role during stakeholder's communication. Modern collaborative tools and digital technology bring two parties in collaboration together that helps in reducing effect of culture variation and language barrier. In one of research study,<sup>3</sup> collaborative tool EGRET was suggested, which aims to support requirements management and favor effective communication during requirements negotiations. Stakeholders and experts from different domains can effectively engaged with clients with EGRET where interfaces were developed from perspective of both clients and vendors. Any change in

requirements can easily be tackled using requirements traceability. Other advantages of using EGRET include supporting informal collaboration of different parties. Another research<sup>4</sup> study presents the Spatial Hypertext Wiki as a collaborative tool for supporting creativity in the RE process. The Spatial Hypertext characteristics of the wiki provide a virtual board where distributed stakeholders can share, brainstorm, negotiate, or prioritize the knowledge involved in RE.

Requirements management during elicitation process is very important specially when dealing with large size requirements in GSD as due to challenges such as lack of effective communication and makes it difficult process. According to Minhas and Zulfikar,<sup>5</sup> we need requirements change management (RCM) frameworks for requirement elicitation to deal with changes in requirements. Significance of managing requirements during elicitation increases more when we talk about GSD because physical face-to-face meetings in stakeholders become difficult. Requirements prioritization and traceability are two important aspects to be considered during requirements management. In one of research study, RCM framework is suggested by Khan et al.<sup>6</sup> for GSD where step-by-step evaluation of change in requirements during elicitation is presented in RCM framework. According to Khan et al.,<sup>6</sup> managing change in requirements is hard specially in GSD due to communication problem, and thus, proper RCM will help to handle changes in requirements and complete the project in time and cost. According to Kumar and Kumar,<sup>7</sup> requirement management in GSD is quite challenging and important because one of the failure reasons of software is that requirements are not properly and well managed during elicitation phase, and for efficient management of requirements, authors proposed ontology-based approach. Requirements traceability is important factor during management of requirements and is considered in this approach. Another requirement management approach is presented for GSD in one of research study<sup>8</sup> that consists of four phases, that is, (1) establishing and maintaining a requirements repository; (2) generating a requirements traceability matrix; (3) communicating and discussing requirements; and (4) RCM. This shows all critical phases of requirement management are covered in this approach.

Fixing errors during latter stages of development become very difficult and costly. We need coaching and trainings of stakeholders involve in GSD: trainings on how to communicate with each other specially across different cultures; training on using collaborative tools, elicitation techniques for GSD, and how to use defined process models suitable for GSD. According to Romero et al.,<sup>9</sup> skills and knowledge of requirements are needed for requirement elicitation in GSD, and thus, role of trainings cannot be underestimated.

Frequent discussion on requirements with clients further clarify confusions and conflicts on requirements.<sup>10</sup> In GSD where effective communication is difficult, clients and vendors need more and more discussions on requirements. As in GSD, traditional elicitation techniques such as background study and apprenticing are difficult; thus, use of effective elicitation techniques suggested for GSD should be used. Using collaborative elicitation methods helps further in proper requirements collection and implementation. Using requirement engineering modeling specific for GSD and collaborative tools is more helpful.

Process maturity and process selection is considered as one of the success factors in GSD.<sup>1</sup> For example, in one of research paper,<sup>11</sup> an iterative model for global requirements elicitation is proposed and evaluated using case study, and the results show that by introducing iteration in the RE process can satisfy customers in satisfying their requirements. In another research study,<sup>12</sup> collaborative elicitation method is adopted based on social intelligence for networked software and analyzes its self-adaptive and emergence feature. Author used requirements semantic concepts, and semantic web technology is adopted for supported distributed requirement elicitation. In another research study,<sup>13</sup> collaborative elicitation with visualization technique is used with involvement of users. Suggested social visualization using bubble chart is maintained by the prototype using Fusion Charts where each stakeholder is represented by a bubble. According to Korkala and Abrahamsson,<sup>14</sup> advance process models such as agile should be used for GSD because it focuses more on communication and negotiations that is very crucial during elicitation specially when stakeholders are distributed.

According to Aranda et al.,<sup>15</sup> knowledge management in GSD is a very big challenge as requirements can come from various stakeholders and sources, and it is more difficult when large size requirements are dealt by various stakeholders across different sites. Knowledge sharing activities should be favored in organizations so that problems and challenges during elicitation can be handled and thus organizational infrastructure improvement practices are necessary. According to Humayun and Gang,<sup>16</sup> shared understanding of requirements is very necessary by all team member. Through experiment, it was analyzed that a clear organizational structure with communicating responsibilities helps to improve shared understanding of requirements in GSD despite there exist severe challenges.

In GSD, developing mutual trust and understanding is necessary. With more and more collaboration in stakeholders, mutual trust and understanding can be developed. No team formation is possible with mutual trust, which is possible only with collaboration. According to Damian and Zowghi,<sup>17</sup> due to lack of physical frequent meetings and culture difference, gaining mutual trust is difficult, and thus with collaboration and informal communication, mutual trust can be gained.

Besides the importance of requirements elicitation in GSD, there is no assessment model yet suggested that will assist software vendors to evaluate their maturity towards requirements elicitation process. Our current research will fulfill this research gap. By implementing an assessment model will help not only software vendors but will also assist clients in selection of vendors and development organizations.

## 2 | AIMS AND OBJECTIVES OF THE RESEARCH

The objective of our research study is to develop REM for GSD vendors. The basic concept, design, and structure of REM are published in our previous paper.<sup>18</sup> In this research article, we have implemented and evaluated REM with case studies. This model will assist GSD vendors in implementation of requirement elicitation phase in GSD. This model will help GSD organization's in identifying, analyzing, and implementing the CSFs of REM with the help of practices provided. The REM will be the end result of two separate SLR studies, an empirical investigation in the industry and two case studies in the relevant organization.

- In order to achieve the above stated objectives, we have planned the following research phases.
- In the first phase, the data will be collected through SLR and empirical study for the identification of CSFs during elicitation in GSD, from vendor's perspectives.
- In the second phase, real-world practices will be identified through SLR and empirical study for implementation of previously identified CSFs.

In the last phase, case study will be conducted for evaluation of the REM in industrial environment.

## 3 | RQs

The following RQs motivated the work reported in this paper:

**RQ1:** What are the critical success factors, as identified in the literature, that assist software vendors organizations in conducting requirement elicitation process in GSD?

**RQ2:** What are the critical success factors, as identified in the real practice, that assist software vendors organizations in conducting requirement elicitation process in GSD?

**RQ3:** What are the solutions/practices, as identified in the literature, for conducting requirement elicitation process in GSD?

**RQ4:** What are the solutions/practices, as identified in real practices apart from literature, for conducting requirement elicitation process in GSD?

**RQ5:** Is the GREM practically robust in terms of measuring organization's (vendor) capability for conducting requirement elicitation process in GSD?

Till date, the findings of RQ1 and RQ2 have been identified and published in our previous research article.<sup>19</sup> The main contribution of this research study is the development of the REM for GSD vendors. The REM provides a practical structure that will help GSD organizations to measure their capability for REM. This study is novel and significant as no previous studies for the identification of success factors of requirements elicitation through SLR can be found in the literature. Similarly for the identification of practices for proper implementation of CSFs of requirements elicitation in GSD from vendor's perspectives, no SLR has been conducted. Our developed model REM is the outcome of two SLRs and empirical industrial studies.

## 4 | BACKGROUND STUDY

Literature review of existing studies on assessment models shows that there exist two perspectives for the design and implementation of an assessment (maturity) model, that is, perspective of potential performance and perspective of life cycle.<sup>20,21</sup> However, most of the recent published research found are on potential performance perspective,<sup>20,21</sup> where an organization may attain advanced stage based on their performance and there is no further improvement needed when organization reaches the "final" maturity stage.<sup>21</sup> Nowadays, most of the models are designed and developed through potential performance perspective<sup>20</sup> and are based on the classic staged maturity framework.

### 4.1 | Approaches to maturity assessment model development

According to Wendler,<sup>21</sup> from literature, it can be found that the authors follow mostly unknown assessment models, whereas some follow widely adopted approach developed by standard organization like, COBIT by CioIndex, QMMG by Business Process Maturity Model (BPMM)<sup>22</sup> developed by Object Management Group (OMG), Software Process Improvement and Capability dEtermination (SPICE)<sup>23</sup> by Software Quality Institute (SQI), ISO/IEC 15504<sup>24</sup> and ISO-33000 family of standards by ISO, and Capability Maturity Model Integration (CMMI) by SEI (Software Engineering Institute).<sup>25</sup> Researchers use different approaches while designing any assessment model. Summary of different assessment models are given below.

### 1. QMMG BY CROSBY'S

This approach uses six features of the quality management with five maturity levels. The levels of QMMG range from certainty (L5), wisdom (L4), enlightenment (L3), and awakening (L2). Maturity models based on this approach are Maier et al.<sup>26</sup> and Brotons Martínez and Sansalvador Selles.<sup>27</sup>

### 2. BPMM BY OMG

This approach provides process improvement and guide manager to monitor business process efficiency. Different levels of this model include Innovating (L5), predictable (L4), standardized (L3), managed (L2), and initial (L1).<sup>27,28</sup>

### 3. SPICE (SOFTWARE) BY SQI

It contains nine parted documents that is used to support software process assessment and improvement.<sup>23</sup> Different levels include continuously improving (L5), quantitatively controlled (L4), well defined (L3), planned and tracked (L2), performed informally (L1), and not performed (L0). Maturity models based on this approach are Simon<sup>29</sup> and Kuvaja.<sup>30</sup>

### 4. ISO/IEC 15504 AND ISO-33000 FAMILY OF STANDARDS

The SPICE uses six maturity levels, that is, optimizing (L5), predictable (L4), established (L3), Managed (L2), performed (L1), and incomplete (L0). Maturity models based on this approach is Garzás et al.<sup>31</sup>

### 5. CMM/CMMI BY SEI

CMMI is a widely adopted organization capability assessment model.<sup>32</sup> CMMI model contains five capability maturity levels (CMLs).<sup>33</sup> The highest CML is level 5 that represents an outstanding capability (a quality production), whereas the lowest CML is 1, which denotes an ailing controlled software engineering process.<sup>25</sup>

## 4.2 | Mini assessment model as an alternative to CMMI

Most of the software organizations adopted software process improvement (SPI) in shape of extensive assessment by software CMMI.<sup>25</sup> However, CMMI-based assessments are costly and time-consuming.<sup>31</sup> According to Staples et al.,<sup>34</sup> several organizations have not adopted CMMI model because of its considerable costs and adopted mini assessment instruments for their easiness.<sup>35</sup> Moreover, CMMI-based assessment is not applicable for some areas of software engineering specifically when we talk about specific areas of software engineering such as requirement elicitation in both local development and GSD.<sup>34</sup>

Example of such mini assessment approach includes Motorola assessment tool<sup>36</sup> that is used to measure progress evaluation of organizations. Motorola assessment tool estimates organizational software development capability and proclaimed the present status of the organization comparative to CMM.<sup>36</sup> Other similar approaches include mini assessment,<sup>34</sup> Dybå scale,<sup>37</sup> and Iskrate weighted system.<sup>38</sup>

Many other models' developer uses the structure of CMMI and develops an assessment model in the domain where CMMI is not directly applicable or it is time-consuming and costly.

Since the first version of the CMMI<sup>25</sup> is released until now, more than 150 assessment models have been published to be used in several fields<sup>39</sup> in software engineering domains such as CML and KPAs, and a number of researchers have developed numerous capability maturity assessments models (CMAMs) for evaluating the organization capability in numerous phenomenon's like SPI implementation,<sup>40,41</sup> outsourcing partnership,<sup>32</sup> requirements engineering,<sup>33,42</sup> software testing,<sup>43</sup> software usability,<sup>44</sup> software process maturity,<sup>33</sup> software quality,<sup>45</sup> software measurement,<sup>46</sup> software workforce,<sup>47</sup> software maintenance,<sup>48</sup> and organizational learning during the software development.<sup>49</sup>

Several authors have modified the structure of CMMI and applied in new study domains such as knowledge sharing management,<sup>50</sup> medical security,<sup>51</sup> digital investigations,<sup>52</sup> digital game,<sup>53</sup> energy efficiency,<sup>54</sup> and analytic maturity.<sup>55</sup>

In adopting and implementing assessment model, several previous studies like previous studies<sup>42,43,45,47-52,56</sup> directly used the complete CMMI structure. Though, several research studies like,<sup>33,40,41,53,57</sup> the KPAs was substituted by critical success factors (CSFs). Existing CMAMs have both theoretical and industrial contribution to the software community for the assessment of a specific organization in different domains. However, there is an intense need of bringing improvement in existing assessments models so that it can be applied for specific domains of software engineering such as requirement engineering.

### 4.3 | Summary of the CMMI bases models

Motorola assessment instrument<sup>36</sup> was firstly designed and implemented by for assessing organization progress. This method was primarily implemented to drive Motorolas to level 2, which include three dimensions in table columns format where all the key practices can be evaluated using scoring criteria from 0 to 10.

Eastman Kodak Company developed Mini-assessment method (MMA) for evaluating company capability relative to CMM that use score between zeros to one and gives equal importance to all key process area. However, lack of review documents process and intensive interviews reduces meticulousness and trustworthiness of the mini-assessment.<sup>35</sup>

Dybå<sup>37</sup> proposed a tool for evaluating the CSFs in SPI. In their instrument, the author designs an evaluation scale for rating the success factor by giving a score from 1 to 5. Frey-Pueko<sup>38</sup> develops a method called SEPAM (Systems Engineering Progress Assessment Method) for Iskrate Telecommunications Systems Developing Company.

Several CMMI-based CMAMs using CSFs concepts can be found from literature such as SPI-IF implementation maturity model (IMM),<sup>40</sup> R-CMM,<sup>42</sup> REMMF,<sup>57</sup> SAMM,<sup>56</sup> MCCM,<sup>46</sup> OS-UMM,<sup>44</sup> TMMi,<sup>43</sup> AiOLOs,<sup>49</sup> DI-CMM,<sup>52</sup> SOPM,<sup>32</sup> SWAM,<sup>47</sup> UQIM,<sup>45</sup> DGMM,<sup>53</sup> SPIIM,<sup>41</sup> EEM,<sup>54</sup> and APMM.<sup>55</sup> A summarized view of the CMAMs is given in Table 1. Discussion in Section 8 is based on this table.

## 5 | RESEARCH METHODOLOGY

The research design is composed of systematic literature review (SLR), questionnaire survey and case study.<sup>40</sup> Details are given as follow:

**Phase#1:** initially, literature was reviewed through two different SLRs (SLR-1 and SLR-2). SLR-1 was conducted for identification of CSFs whereas SLR-2 was conducted for the identification of practices.

**Phase#2:** questionnaire survey was carried out, in order to identify practices from real practice and to validate the SLRs findings and to pinpoint any other factor/practice apart from the identified ones.

**Phase#3:** development of the proposed model based on the inputs from the SLRs and questionnaire survey.

**Phase#4:** a case study was executed, in order to evaluate the REM in real word environment.

### 5.1 | SLR

For the present study, we have executed two different SLRs, one for finding CSFs and two for finding practices/solutions for proper implementation of these CSFs. SLR process is used<sup>58</sup> as the main methodology for data gathering. To find relevant data on the basis of RQs, the SLR helps to investigate and assess it by using the methodical way from the primary studies. First of all, an SLR plan was transcribed in the form of protocol to describe the strategy for the study. According to Afzal et al.,<sup>59</sup> the three main stages of a SLR process are planning, conducting, and reporting. Design of SLR protocol for success factors identification was published in our previous paper.<sup>19</sup> As a result of execution of SLR protocol, we had set of 17 CSFs from 70 papers as shown in Table 2. Design of SLR protocol for identification of practices for each success factor is published in our previous research paper.<sup>60</sup> However, we have executed it again as last 3 years, papers are now included, and as result, 80 practices were retrieved from 41 papers as shown in Tables B1–B14 of the Appendix B, respectively.

#### 5.1.1 | Data collection via SLR

SLR protocol is designed for the identification of practices for CSFs of requirements elicitation in GSD by following previous research.<sup>61–65</sup> The step-by-step approach of conducting SLR is shown in different phases of Figure 1.

##### *Phase 1: Planning the review*

In this phase, RQs are finalized before conducting review. Based on identified RQs, papers will be retrieved from different digital sources and then inclusion and exclusion criteria will be applied accordingly. From selected list of papers, all possible practices will be identified. The data sources identified are listed in Table 3.

Search strings are formulated from the key words of RQ. Four key words are retrieved from RQ1 as shown in Table 4. For every keyword, some alternative words are also selected because in not all papers, same key words are used by authors. Two operators “AND” and “OR” will be used to make search strings from keywords and alternative words. Operator AND will be used to combine keywords, and operator OR will be used to concatenate different alternative for same keyword.

**TABLE 1** Summary of the CMMI bases capability assessment models.

| S.NO | Year | Name     | Purpose                                              | Data collection                   | No of levels | Validations             |       |                     |    |
|------|------|----------|------------------------------------------------------|-----------------------------------|--------------|-------------------------|-------|---------------------|----|
|      |      |          |                                                      |                                   |              | Technique applied       | Score | Method              | No |
| 01   | 2005 | SPI-IF   | Implementing SPI                                     | Literature + Interviews           | 04           | Motorola                | Crisp | Case study          | 02 |
| 02   | 2005 | R-CMM    | Requirement process validation from experts          | Questionnaire                     | 05           | Dybå scale              | Crisp | Expert panel review | 27 |
| 03   | 2007 | REMMF    | Requirements engineering process maturity measure    | Interviews                        | 03           | Motorola                | Crisp | Case study          | 02 |
| 04   | 2008 | SAMM     | Evaluation of software assurance progress maturity   | Project based                     | 03           | Own design              | Fuzzy | Expert panel review | 33 |
| 05   | 2010 | SOVRM    | Vendors' readiness assessment in outsourcing         | Questionnaire SLR + Questionnaire | 04           | Motorola                | Crisp | Case Study          | 03 |
| 06   | 2010 | MCMM     | Enhancement of company maturity measurement          | Questionnaire                     | 05           | GQM                     | Crisp | Case study          | 01 |
| 07   | 2012 | OS-UMM   | Examine usability open-source projects               | Questionnaire                     | 04           | BOOTSTRAP               | Fuzzy | Case study          | 02 |
| 8    | 2012 | TMMi     | Test maturity model for non-profit organizations     | Case Study                        | 05           | Own design              | Crisp | Case study          | -  |
| 9    | 2013 | AiOLoS   | To assessing the level of organizational learning    | Case Study                        | 03           | GQM                     | Fuzzy | Case-study          | 02 |
| 10   | 2013 | DI-CMM   | Organization examination capability measure          | Interview                         | 05           | Own design              | Crisp | Case Study          | 03 |
| 11   | 2014 | SOPM     | Organizations capacity of outsourcing                | SLR + Questionnaire               | 05           | Motorola                | Crisp | Case study          | 02 |
| 12   | 2015 | SWAM     | Workforce skill valuation                            | SW-CMM                            | NA           | Bloom's scale           | Crisp | Expert panel review | 11 |
| 13   | 2016 | CIP-UQIM | Enhance software quality in SMEs                     | I Questionnaire                   | 05           | Own design              | Crisp | Expert panel review | 17 |
| 14   | 2016 | DGMM     | To improve efficiency of game development projects   | Questionnaire                     | 05           | BOOTSTRAP               | Fuzzy | Case study          | 02 |
| 15   | 2017 | SPIIMM   | Implementation of SPI                                | SLR + Questionnaire               | 05           | Motorola                | Crisp | Case study          | 03 |
| 16   | 2017 | EEM      | Assessment of the contemporary maturity              | Literature survey                 | 05           | Zero defect zero effect | Crisp | Case study          |    |
| 17   | 2018 | APMM     | Evaluation of systematic maturity of an organization | Literature survey                 | 05           | Own design              | Crisp | Not validated       | -  |

Abbreviation: CCMI, Capability Maturity Model Integration.

From Table 4, the finalized search strings used for this research are as follows: (A1) AND (A2) AND (A3) AND (A4).

The finalized search string is given below:

("Practices" OR "solutions" OR "strategies" OR "facilitators" OR "mitigation", "overcome issues" OR "safeguards") AND ("Factors" OR "Elements" OR "characteristics") AND ("Requirement Elicitation" OR "Requirement gathering" OR "requirement engineering" OR "requirements implementation" OR "requirements analysis") AND ("Global Software Development" OR "Multisite Software Development" OR "Offshore Software Development").

To include inappropriate research papers from the total selected literature, the following criteria will be used. During literature selection, not all but some publications that are most relevant to RQ are selected for final data retrieval. Inclusion and exclusion criteria for this research study are based on RQ1. Papers that address practices for RCM during elicitation in GSD will be included. Based on RQ1, inclusion criteria are listed as follows:

**TABLE 2** REM levels.

| Level no | REM level        | CSFs                                                                                                                                                                          | Practices                                                                                                                                                                            |
|----------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Initial contract | Nil                                                                                                                                                                           | Nil                                                                                                                                                                                  |
| 2        | Success          | CSF1: effective and timely communication<br>CSF2: frequent discussions<br>CSF3: using proper elicitation technique for GSD<br>CSF4: requirement management during elicitation | CSF1: practices are presented in Table B1<br>CSF2: practices are presented in Table B2<br>CSF3: practices are presented in Table B3<br>CSF4: practices are presented in Table B4     |
| 3        | Readiness        | CSF5: mutual trust and understanding<br>CSF6: coordination, cooperation, and collaboration practices adopted<br>CSF7: culture awareness                                       | CSF5: practices are presented in Table B5<br>CSF6: practices are presented in Table B6<br>CSF7: practices are presented in Table B7                                                  |
| 4        | Conversion       | CSF8: coaching and trainings<br>CSF9: using collaborative tools and innovative technologies<br>CSF10: RE process maturity<br>CSF11: agents and communication coordinator      | CSF8: practices are presented in Table B8<br>CSF9: practices are presented in Table B9<br>CSF10: practices are presented in Table B10<br>CSF11: practices are presented in Table B11 |
| 5        | Maturity         | CSF12: organizational infrastructure improvement practice<br>CSF13: knowledge sharing and management<br>CSF14: effective global project management                            | CSF12: practices are presented in Table B12<br>CSF13: practices are presented in Table B13<br>CSF14: practices are presented in Table B14                                            |

Abbreviations: CSFs, critical success factors; GSD, global software development; REM, requirement elicitation model.

- Papers written in English language will be selected.
- Source for every paper should be either journal, conference, or book chapter.
- Papers that deal with requirements gathering within the context of GSD will be selected.
- Papers that are available in full text will be selected.

To exclude inappropriate research papers from the total selected literature, the following criteria will be used:

- Papers with the number of pages less than five will be excluded.
- Papers that are although related to GSD but does not concern with requirements elicitation will be excluded.
- Papers that do not include any practice of CSFs of requirement elicitation in GSD context will be excluded.
- All review or SLR papers will be excluded.

#### *Phase 2: Conducting the review*

The second phase in the SLR is conducting the review. This phase consists of two activities, which are research articles selection and data extraction and synthesis.

##### *a. Research articles selection*

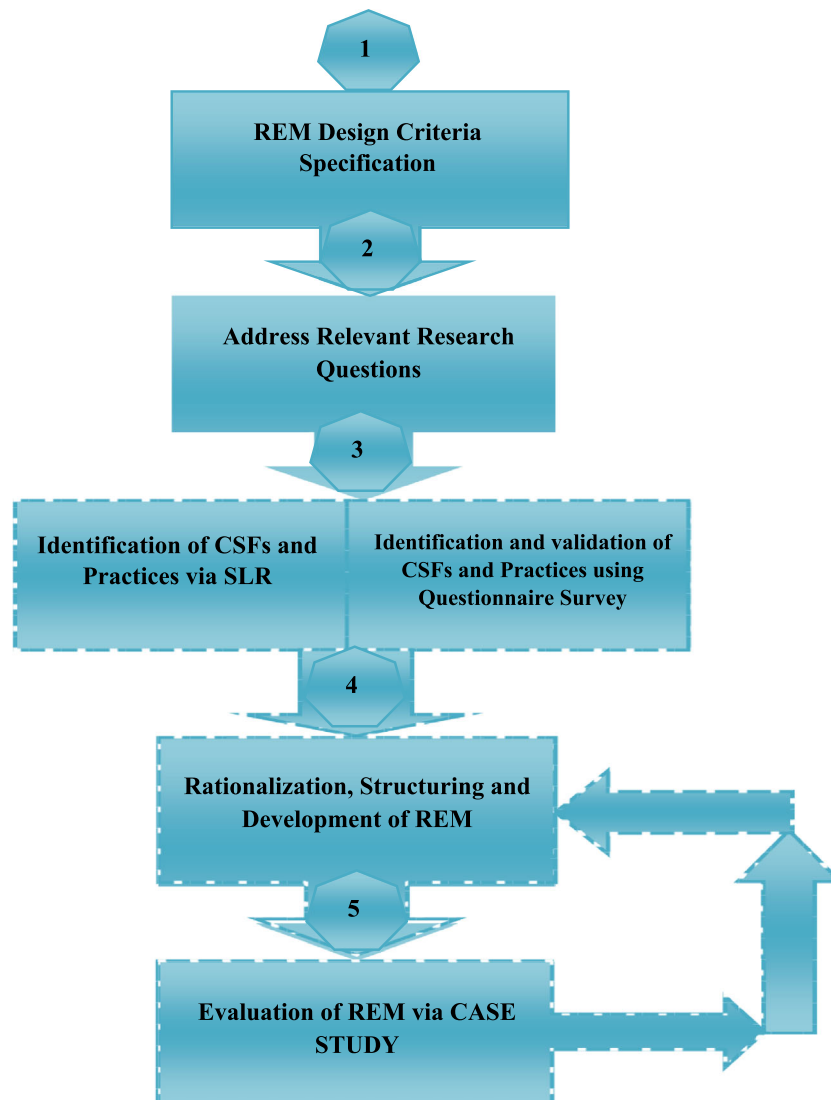
The collection of selected research articles was refined by applying a tollgate method by Afzal et al.<sup>62</sup> Using this method, 432 research articles were selected from the selected collection based on search strings. Next, 41 primary studies were finally selected based on the inclusion and exclusion set as shown in Appendix A. Every selected primary study was labeled as [S] to specify it as the SLR article only. Figure 2 shows final selected papers from digital sources with tollgate approach, where  $p$  represents the papers selected in five phases from different digital sources as result of tollgate approach. This corresponds to 9.5% from the total extracted studies. The list of the final articles selected is given in Appendix A.

##### *b. Data extraction and synthesis*

Data extraction is defined as a process to examine the retrieved information from the selected studies. For each research article, the following data were extracted by using the Microsoft Word Sheet:

The selected research articles were then reviewed to extract the concepts, contribution, or findings. The extracted data were summarized in a compact statement to form the list of practices. During the process of data extraction, four reviewers were involved in order to reduce the researcher's bias and to increase the validity of the practices identified. When the data extraction process has been completed, the inter-rater reliability test was conducted in order to eliminate bias in a personal opinion.





**FIGURE 1** REM development cycle activities.

**TABLE 3** Data sources.

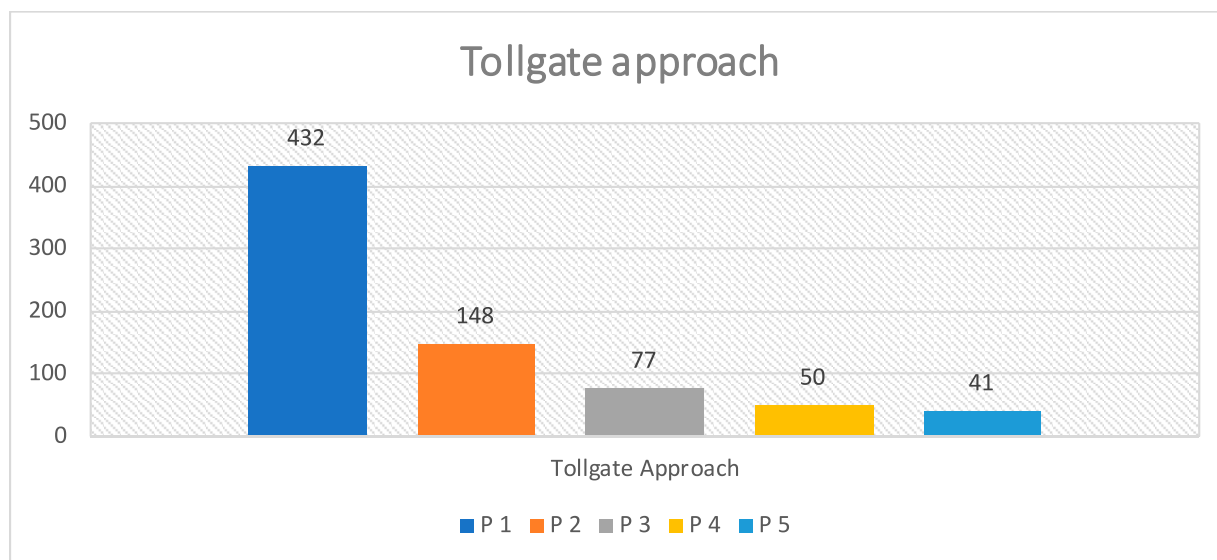
|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electronic databases | Wiley Inter Science ( <a href="http://www.wiley.com">www.wiley.com</a> ).<br>ACM Digital Library ( <a href="http://dl.acm.org">http://dl.acm.org</a> ).<br>Science Direct ( <a href="http://www.sciencedirect.com">www.sciencedirect.com</a> ).<br>Springer Link ( <a href="http://link.springer.com">link.springer.com</a> ).<br>Google Scholar ( <a href="http://scholar.google.com">scholar.google.com</a> ).<br>IEEE Xplore ( <a href="http://ieeexplore.ieee.org">http://ieeexplore.ieee.org</a> ). |
| Search items         | Conference and journal papers in full text                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Language             | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Publication period   | No bound on time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Space                | No limit on number of papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

The external reviewers selected 15 studies from the initial phase (p-1) from the tollgate approach<sup>19</sup> and carried out all phases of SLR process. Non-parametric Kendall's coefficient of concordance ( $W$ ) was used in order to check the inter-rater agreement among the reviewers. The value of  $W = 0$  indicate a complete disagreement among the reviewers, whereas  $W = 1$  indicates a complete agreement.



**TABLE 4** Search string used in research.

| Keywords                         | Alternatives                                                                                                 |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Requirements elicitation (A1)    | ("Requirement gathering," "requirement engineering," "requirements implementation," "requirements analysis") |
| Global software development (A2) | ("Multisite Software Development," "Offshore Software Development")                                          |
| Factors (A3)                     | ("Elements," "characteristics")                                                                              |
| Practices (A4)                   | ("solutions," "strategies," "facilitators," "exercising," "overcome issues," "safeguards")                   |

**FIGURE 2** Tollgate approach.

### Phase 3: Reporting the review

Phase 3 of the SLR is reporting the review. In this phase, practices for CSFs are identified. Next, the distribution of the selected studies along with the research methods applied in the studies are reported.

## 5.1.2 | Empirical study

We had conducted questionnaire survey in two phases. In phase 1, responders were asked to add new practices for each success factor that were not previously identified via SLR, and in phase 2, responders were asked to give their valuable feedback on the identified practices via SLR and questionnaire survey of phase 1.

For the validation of our SLRs findings, we have conducted questionnaire survey in industry from total of 35 experts of GSD. Empirical survey is a method, for obtaining a numeric description on the sample. It is the most widely used research methodology used for data collection, in order to obtain tacit information on a particular phenomenon or problem of interest.<sup>66</sup>

Design of questionnaire is based on the input of SLR findings. Demographic information is the first section. Section 2 presents a list of 14 success factors that are evaluated on a seven argument Likert scale, that is, 1—*Extremely Agree* (EA), 2—*Moderately Agree* (MA), 3—*Slightly Agree* (SA), 4—*Not Sure* (NS), 5—*Extremely Disagree* (EDA), 6—*Moderately Disagree* (MDA), and 7—*Slightly Disagree* (SDA).

The questionnaire procedures were tested with the help of five experts in domain of software engineering of Pakistan.

Invitation letter was written for responders briefing on questionnaire and were mailed on Yahoo (<https://groups.yahoo.com/neo>), LinkedIn ([www.linkedin.com](http://www.linkedin.com)), and Facebook ([www.facebook.com](http://www.facebook.com)).

In comeback to these emails, more than 70 industry professionals' experts agreed for contribution. We received 45 filled questionnaires. Applying the quality criteria, 10 questionnaires were dropped/excluded. A final of 35 responses remain for further analysis. Among the 35 participants, 12 experts are foreigners, whereas the remaining 23 experts are Pakistani nationals. Further analysis of data was made based on these 35 experts' opinion.

We have conducted *t* tests for checking non-response biasness cause and effects. We did not find any important dissimilarity ( $p < 0.05$ ) among non-respondents and respondents.<sup>30</sup>

## 5.2 | REM development

### 5.2.1 | REM development process

We have followed the concept of CMMI, IMM, and SOVRM<sup>40,67,68</sup> and SOPM<sup>32</sup> perspectives and developed our REM model as shown in Figure 1. The proposal of our REM model has been published.<sup>18</sup> REM is composed of five levels. The identified success factors along with their practices were distributed across the five levels of the model. These levels and distribution of the factors/practices were presented to software engineering research group, and necessary changes were made as suggested. It was further reviewed through external/foreign reviewer, and necessary changes were made accordingly. Finally, the model was evaluated through two case studies in the industry. We got agreement of all responders from questionnaire on these five levels of REM. The sated model was developed using five research stages. In the first stage, criteria for REM were developed. Step-by-step design and implementation of REM are shown in Figure 1.

### 5.2.2 | REM evaluation

The case study method will be used to evaluate the REM as it is considered a powerful evaluation tool and can provide useful real-world information.<sup>69</sup> Two case studies will be conducted where REM will be used to evaluate its effectiveness in software development of GSD. At the end of each case study, focus group sessions<sup>70</sup> will be conducted with the participants in order to obtain feedback about the REM.

### 5.2.3 | Structure of the REM

We will build the structure of the REM on the following three dimensions:

- REM levels
- Factors (CSFs) in each level
- Practices for implementation of CSFs

Int this research, we have adapted structure CMMI, the IMM, SOPM<sup>32</sup> and Software Outsourcing Vendor Readiness Model (SOVRM) for designing REM in order to assist GSD organizations in better collection of requirements and assess their weakness and strengths. REM levels are finalized from already published models in this domain while success factors and practices for each success factor as identified from SLR are mapped to every level then accordingly. For gaining confidence on the structure of REM, feedback from industrial experts is considered.

## 6 | RESULTS

Results based on RQ1 and RQ2 are published in our previous research article. Results based on RQ3 and RQ4 are presented below.

### 6.1 | Practices identified via SLR (RQ3)

The identified practices as a result of SLR from 41 research papers are given in Tables B1–B14 of Appendix B. Column “identified via” shows which of these practices are identified via SLR from these 41 articles. As a result, we have total of 80 practices for all success factors via SLR.

### 6.2 | Practices identified from real practice (RQ4)

The identified practices as a result of questionnaire survey of phase 1 are given in Tables B1–B14 of Appendix B. Column “identified via” shows which of these practices are identified via questionnaire. We had total of 35 practices that are identified via questionnaire.

### 6.3 | REM model and evaluation (RQ5)

Thus, from both SLR and questionnaire, total number of practices become 115. These 115 practices were then validated via questionnaire in phase 2.

We have identified a total of 17 success factors in which 14 was marked as CSFs. For the proper implementation of these CSFs, we have identified 115 practices through SLR and empirical survey in the industry. The classification of CSFs in different categories had led us to design different levels of the REM for GSD vendors. Each level of the REM has different CSFs. Under each CSF, numerous practices have designed that will guide GSD vendor organizations in implementing that CSF. For an organization to achieve certain REM level, they should address each CSF under that particular level. Most of the REM levels (except level 1) comprise different CSFs. Each CSF contains different REM practices.

**Initial contract:** the first level is “Initial contract.” This level can be defined as one of chaotic processes. Here, as a result of project kickoff meeting, elicitation process starts.

**Success:** the second level in REM is “Successful implementation.” The focus of this level is on basic needs and pre-requisites of elicitation such that vendors must have to implement these CSFs during requirements elicitation. If company fails to achieve level 1 or these factors, this shows elicitation process is not conducted. This level has four CSFs and its corresponding practices as shown in Tables B1–B4 of Appendix B.

**Readiness:** the third level of REM is “elicitation readiness.” Trust of clients has been gained, and both vendors and clients have achieved collaboration and culture awareness of each other's. Without collaboration, trust, and culture awareness, it might be possible that organizations have collected clients' requirements as in level 2, but readiness level shows more successful towards maturity of organizations to meet future challenges that might occur or less prone to errors in requirements in review phase. This level has three CSFs and its corresponding practices as shown in Tables B5–B7 of Appendix B.

**Conversion:** this is the fourth level of REM towards final maturity level. Here, organizations have completely implemented collaborative tools, facilitating trainings sessions, showing maturity in apply RE process models, and using agents and communication coordinators during conversation with clients. Conversion and implementation have been successfully done at this level. This level contains 4 CSFs, and its corresponding practices are shown in Tables B8–B11 of Appendix B.

**Maturity:** the fifth and final REM level is “Maturing in elicitation.” This level emphasizes more on management, organizational infrastructure, and knowledge sharing management. This level contains 3 CSFs, and its corresponding practices are shown in Tables B12–B14 of Appendix B.

Different studies have confirmed the value of the CSF approach in different fields.<sup>68,71,72</sup> Therefore, the CSF approach may also be effective in GSD activities.

Like CMMI PAs, CSFs and its practices were distributed into different levels. A list of practices designed for each CSF (included in REM levels) has been provided in Appendix B.

These practices were identified via surveys with 50 participants. All the practices were included in the final list as practitioners have identified these practices based on their real-life outsourcing experiences. However, a thorough review was conducted in order to remove the duplicate practices.

For the evaluation of our planned model REM, we have used Motorola assessment tool<sup>36</sup> as revealed in Table 5.

The same instrument is also used for the evaluation of SOVRM. This tool has also been used in CMMI to evaluate the organization's present position comparative to Capability Maturity Model (CMM) and to recognize weak zones that required improvements and management attention.<sup>36</sup>

The Motorola's instrument is proposed by Daskalantonkis Motoroh.<sup>36</sup> The following are the three evaluation dimensions of the Motorola assessment instrument:

**Approach:** criterion for this dimension is the management support and organization's commitment for the practice as well as the organization's capability to address the practice.

**Deployment:** criterion for this dimension is the fullness and consistency of practice implementation through every project region.

**Results:** Breadth and consistency of positive outcomes over time and throughout project areas are the criterion. A score from 0 to 10 is given for every dimension.

Score value from 0 to 10 will be assigned to each dimension, which can be decided based on the guidelines given in Table 5.

The following steps have been adapted for assessment component using the Motorola instrument. One example is presented in Table 6.

- Step one: compute the three-dimensional scores of the Motorola evaluation tool for each practice.
- Step two: sum the three-dimensional scores for each practice, then divide the sum by three and round to next-door whole number.
- Step three: replicate this process for each and every practice. Sum totals the score of each practice and find average gain an overall total score for that particular factor.
- Step four: connecting the assessment scores to REM levels: a score of seven or greater for particular CSF will direct that, that individual factor has been effectively addressed. Any CSFs that have an average score below seven is considered weak.

**TABLE 5** Motorola evaluation tool.

| Key activity evaluation dimensions |                                                                                                                                                                                                       |                                                                                                                                                                                                                |                                                                                                                                                                                  |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score                              | Approach<br>0, 2, 4, 6, 8, 10                                                                                                                                                                         | Deployment<br>0, 2, 4, 6, 8, 10                                                                                                                                                                                | Results 0, 2, 4,<br>6, 8, 10                                                                                                                                                     |
| Poor—0                             | <ul style="list-style-type: none"> <li>No organizational ability (OR)</li> <li>Practice either not evident or no management appreciation of necessity</li> </ul>                                      | No part of the organization express interest or uses the practice                                                                                                                                              | ✓ Ineffective                                                                                                                                                                    |
| Weak—2                             | <ul style="list-style-type: none"> <li>Management starts to know the importance and use (OR)</li> <li>A few parts of organization are able to implement the practice</li> </ul>                       | Deployed in some parts of the organization (OR)<br>Inconsistent use                                                                                                                                            | ✓ Inconsistent results (OR)<br>A sign of efficacy for some parts of the organization                                                                                             |
| Fair—4                             | <ul style="list-style-type: none"> <li>Road map for practice execution defined (OR)</li> </ul>                                                                                                        | Some uniformity in use (OR)<br>Less fragmented use (OR) Used in some major parts of the organization                                                                                                           | ✓ Unpredictable results for other parts of the organization (OR)<br>✓ Reliable and positive results for several parts of the organization                                        |
| Marginally qualified—6             | <ul style="list-style-type: none"> <li>Practice implementation across every part of the organization (OR)</li> </ul>                                                                                  | Used in many parts of the organization (OR)<br>Mostly stable across various parts of the organization                                                                                                          | ✓ Positive quantifiable results in maximum parts of the organization                                                                                                             |
| Qualified—8                        | <ul style="list-style-type: none"> <li>Practice established as an essential part of the procedure</li> </ul>                                                                                          | Constant use across nearly every part of the organization (OR)                                                                                                                                                 | ✓ Constantly positive outcomes with passage of time across nearly every part of the organization (OR)<br>✓ Positive measurable outcomes in nearly every part of the organization |
| Outstanding—10                     | <ul style="list-style-type: none"> <li>Organizational excellence in the practice recognized even outdoor the firm (OR)</li> <li>Management provides enthusiastic leadership and commitment</li> </ul> | Verification/monitoring for every part of the organization (OR)<br>Consistent use with the passage of time throughout the organization (OR)<br>Universal and constant set up in every part of the organization | Constantly world class results (OR)<br>Guidance sought by others                                                                                                                 |

**TABLE 6** CSFs evaluation example (CSF1: effective communication).

| CSF1:                                                                                          |                                                               |                                    |                            |                         |                                                       |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------|----------------------------|-------------------------|-------------------------------------------------------|
| S No                                                                                           | Practices                                                     | Key Activity evaluation dimensions |                            |                         | Average score (average of the three dimension values) |
|                                                                                                |                                                               | Approach<br>0,2,4,6,8,10           | Deployment<br>0,2,4,6,8,10 | Results<br>0,2,4,6,8,10 |                                                       |
| 1                                                                                              | Practice asynchronous communication methods                   | 10                                 | 8                          | 6                       | 8                                                     |
| 2                                                                                              | Practice synchronous communication methods                    | 8                                  | 8                          | 6                       | 7                                                     |
| 3                                                                                              | Frequent and open communication with clients on regular basis | 8                                  | 8                          | 8                       | 8                                                     |
| 4                                                                                              | Implement collaborative tools                                 | 6                                  | 6                          | 6                       | 6                                                     |
| 5                                                                                              | Use of semantic web technology                                | 0                                  | 0                          | 0                       | 0                                                     |
| Sum of average scores = (Add all values in the right most column of the table)                 |                                                               |                                    |                            |                         | 29                                                    |
| Overall Score: (Dividing “Sum of average scores” by “total no. of practices”) = $(29/5 = 5.8)$ |                                                               |                                    |                            |                         |                                                       |

- Step five: to attain any REM level by a vendor organization, it is necessary that all CSFs that belong to that REM level must have an average score greater than seven. For example, to attain the REM level 2, it is significant that all of the following CSFs that belong to level 2, that is, “Effective and timely communication,” “Frequent discussions,” “using proper elicitation technique for GSD,” and “requirement management during elicitation,” must have an average score of seven or above.

The evaluation process for any of the CSF is explained with the following example.

## 7 | EVALUATION OF REM VIA CASE STUDY

In order to evaluate REM, two case studies were undertaken in software development companies that are involved in GSD. Case study was adopted as it provides enough information and in-depth analysis for evaluating any assessment model.<sup>32</sup> Because the REM is designed for real industrial practitioners, the case study research method is considered for validation. Case studies were conducted in two different companies. Companies were selected for case studies because they provided particularly rich descriptions of their GSD activities. These two software companies were registered with Pakistan Software Export Board. Training session of 1 day was held in these organizations on evaluation process of REM. The assessment results of each company are presented at Tables 7 and 8, respectively. Each participant of the case study was informed to do assessment based on observations and experiences of previous projects. At the end of case study, results were collected from focal persons of each company on REM evaluation. Detail of each case study in both companies with results is discussed in subsections below.

### 7.1 | Case study at Company A

Company A (ERISP) is a software developing organization located in Peshawar Pakistan ([www.erisp.net](http://www.erisp.net)). This company works on ODOO ERP systems and has more than 40 clients across Pakistan and different countries of Asia. Table 7 shows that Company A stands at level 3 of REM. According to this assessment, a score of 7 or higher for each CSF will indicate that specific factors have been successfully implemented. Any CSFs with an average score that falls below seven are considered a weakness. In order to achieve level 4 of REM, company should implement practices for success factors of level 4 i.e. Conversion and Maturity.

### 7.2 | Company B

Company B is a nationwide company (ISO 9001: 2000 certified) that provides consultancy and information technology services to clients both in the private and public sector, employing almost 1400 professionals as well as having a presence in all major cities of the country. Table 8 demonstrates the evaluation outcomes of Company B. Table 8 shows that Company B positions at level 4, that is, CONVERSION level of the REM. In

**TABLE 7** Implementation scores for each CSFs for the Company A.

| REM levels |            |                                                                      | Company B |                 |
|------------|------------|----------------------------------------------------------------------|-----------|-----------------|
| Level no   | Name       | Critical success factors                                             | Score     | Status          |
| 1          | CONTRACT   | Nil                                                                  |           |                 |
| 2          | SUCCESS    | CSF1: effective and timely communication                             | 8         | Implemented     |
|            |            | CSF2: frequent discussions                                           | 8         | Implemented     |
|            |            | CSF3: using proper elicitation technique for GSD                     | 8         | Implemented     |
|            |            | CSF4: requirement management during elicitation                      | 8         | Implemented     |
| 3          | READINESS  | CSF5: mutual trust and understanding                                 | 8         | Implemented     |
|            |            | CSF6: coordination, cooperation, and collaboration practices adopted | 8         | Implemented     |
|            |            | CSF7: culture awareness                                              | 8         | Not implemented |
| 4          | CONVERSION | CSF8: coaching and trainings                                         | 6         | Not implemented |
|            |            | CSF9: using collaborative tools and innovative technologies          | 5         | Not implemented |
|            |            | CSF10: RE process maturity                                           | 7         | Not implemented |
|            |            | CSF11: agents and communication coordinator                          | 4         | Not implemented |
| 5          | MATURITY   | CSF12: organizational infrastructure Improvement practice            | 5         | Not implemented |
|            |            | CSF13: knowledge sharing and management                              | 6         | Not implemented |
|            |            | CSF14: effective global project management                           | 6         | Not implemented |

Abbreviations: CSFs, critical success factors; GSD, global software development; REM, requirement elicitation model.

**TABLE 8** Implementation scores for each CSFs for the Company B.

| REM levels |            |                                                                      | Company B |                 |
|------------|------------|----------------------------------------------------------------------|-----------|-----------------|
| Level no   | Name       | Critical success factors                                             | Score     | Status          |
| 1          | CONTRACT   | Nil                                                                  |           |                 |
| 2          | SUCCESS    | CSF1: effective and timely communication                             | 9         | Implemented     |
|            |            | CSF2: frequent discussions                                           | 9         | Implemented     |
|            |            | CSF3: using proper elicitation technique for GSD                     | 8         | Implemented     |
|            |            | CSF4: requirement management during elicitation                      | 8         | Implemented     |
| 3          | READINESS  | CSF5: mutual trust and understanding                                 | 8         | Implemented     |
|            |            | CSF6: coordination, cooperation, and collaboration practices adopted | 8         | Implemented     |
|            |            | CSF7: culture awareness                                              | 8         | Implemented     |
| 4          | CONVERSION | CSF8: coaching and trainings                                         | 8         | Implemented     |
|            |            | CSF9: using collaborative tools and innovative technologies          | 8         | Implemented     |
|            |            | CSF10: RE process maturity                                           | 8         | Implemented     |
|            |            | CSF11: agents and communication coordinator                          | 8         | Implemented     |
| 5          | MATURITY   | CSF12: organizational infrastructure Improvement practice            | 5         | Not implemented |
|            |            | CSF13: knowledge sharing and management                              | 6         | Not implemented |
|            |            | CSF14: effective global project management                           | 6         | Not Implemented |

Abbreviations: CSFs, critical success factors; GSD, global software development; REM, requirement elicitation model.

order to attain maturity level 5 of REM, company must have to implement identified practices of success factors “Organizational infrastructure Improvement practice,” “knowledge sharing and management,” and “Effective global project management.”

## 8 | CONCLUSION AND FUTURE WORK

By reviewing the relevant literature through SLR1, we have identified a list of 17 success factors of requirements elicitation in GSD. From the list of these 17 factors, 14 CSFs were finalized for further study based on experts opinion.

For the identified 14 CSFs, we have identified 115 practices using another SLR and questionnaire survey, tagged as SLR2. Based on the inputs from the SLRs and industrial survey, the REM was framed using the CMMI and IMM approach. The 14 CSFs and 115 practices are distributed into five levels. For the evaluation of REM, we have conducted two case studies at two vendor organizations, coded as Company A and Company B. The results of case studies also indicate that if company is on higher level, then it must achieve all its low levels. Take an example of Company B; it is on level 4 so Company B has already achieved levels 1–3 and fails to achieve higher level 5. This also indicates that if company has achieved the highest maturity level of REM then low levels are also achieved. Besides implementing practices of communication and adopting suitable requirements elicitation techniques, companies should work on their infrastructure and global project management. If a company achieves all practices for the success factors such as “Organizational infrastructure Improvement practice,” “knowledge sharing and management,” and “Effective global project management” with average score above 7, then it is obvious that a company has attained all low levels of REM.

In the future, we plan to further elaborate these identified 14 CSFs, and more practices will be identified for these success factors. In the future, we also aim to conduct more case studies on user satisfaction and to make it more improved.

Furthermore, we will automate the REM in the form of a software tool. This tool will generate different assessment reports and will perform different activities related to REM and will guide the GSD practitioners in successfully assessing the organization's level concerning CSFs. From the results of the current study, we have identified a number of further gaps in this area that we plan to do in the future. These include as follows:

- Identifying status of the CSFs implementation, that is, weak and strong, in small-, medium-, and large-sized organizations.
- Identification of activities involved in each level of the REM.
- Identification of risks in each level of the REM.
- Identification of the CSFs from client's perspective.



## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the supplementary material of this article.

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**How to cite this article:** Yaseen M. Requirement elicitation model for global software development vendors. *J Softw Evol Proc*. 2024;36(6): e2628. doi:[10.1002/smr.2628](https://doi.org/10.1002/smr.2628)

## APPENDIX A: LIST OF SLR PAPERS FOR PRACTICES IDENTIFICATION

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## APPENDIX B: PRACTICES FOR CRITICAL SUCCESS FACTORS (CSFs) FOR REM

**TABLE B1** Practices for implementing "effective and timely communication."

| Practice no | Practices                                                     | Identified via |
|-------------|---------------------------------------------------------------|----------------|
| CSFP-1      | Practice asynchronous communication methods                   | SLR            |
| CSFP-2      | Practice synchronous communication methods                    | SLR            |
| CSFP-3      | Frequent and open communication with clients on regular basis | SLR            |
| CSFP-4      | Implement collaborative tools                                 | SLR            |
| CSFP-5      | Use of semantic web technology                                | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B2** “Frequent discussions on requirements.”

| Practice no | Practices                                             | Identified via |
|-------------|-------------------------------------------------------|----------------|
| CSFP-1      | Management support on frequent discussion is required | SLR            |
| CSFP-2      | Negotiate with right person                           | SLR            |
| CSFP-3      | More and more collaboration and coordination          | SLR            |
| CSFP-4      | Focus on interests rather than positions              | SLR            |
| CSFP-5      | Use of hard copy documents during discussion          | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B3** Practices for implementing “Using proper elicitation technique for GSD.”

| Practice no | Practices                                                       | Identified via |
|-------------|-----------------------------------------------------------------|----------------|
| CSFP-1      | Use scenarios to describe the interactions of system and users. | SLR            |
| CSFP-2      | Use prototyping during elicitation                              | SLR            |
| CSFP-3      | Selection of an official language during requirement            | SLR            |
| CSFP-4      | Requirements workshops should be arranged                       | SLR            |
| CSFP-5      | Using collaborative techniques and tools                        | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B4** Practices for implementing “Requirement management during elicitation.”

| Practice no | Practices                                                                                  | Identified via |
|-------------|--------------------------------------------------------------------------------------------|----------------|
| CSFP-1      | Traceability for change management                                                         | SLR            |
| CSFP-2      | Knowledge of interaction between requirements and architecture                             | SLR            |
| CSFP-3      | Each change on a requirement shall be notified to the analysts                             | SLR            |
| CSFP-4      | Usage requirements management tools (database)                                             | SLR            |
| CSFP-5      | Identifying the volatile requirements                                                      | SLR            |
| CSFP-6      | Recording of the rejected requirements                                                     | SLR            |
| CSFP-7      | Requirement change management (RCM) frameworks are needed                                  | SLR            |
| CSFP-8      | Apply proper requirements prioritization techniques (organize and prioritize requirements) | SLR            |
| CSFP-9      | Use proper documents formats                                                               | SLR            |
| CSFP-10     | Define acceptance criteria for requirements selection                                      | SLR            |
| CSFP-11     | Elicit requirements from different resources                                               | SLR            |
| CSFP-12     | Make sure all parties or stakeholders are in agreement on requirements                     | SLR            |
| CSFP-13     | Prepare requirements for verification                                                      | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B5** Practices for implementing “Mutual Trust and understanding.”

| Practice no | Practices                           | Identified via |
|-------------|-------------------------------------|----------------|
| CSFP-1      | Schedule ongoing informal meetings  | SLR            |
| CSFP-2      | Try to build personal relationships | SLR            |
| CSFP-3      | Face-to-face meetings               | SLR            |
| CSFP-4      | Cultural understanding              | SLR            |
| CSFP-5      | More and more collaboration         | SLR            |
| CSFP-6      | Local workshops                     | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B6** Practices for implementing “Coordination, cooperation, and collaboration practices adopted.”

| Practice no | Practices                                                  | Identified via |
|-------------|------------------------------------------------------------|----------------|
| CSFP-1      | Using collaborative tools and technologies                 | SLR            |
| CSFP-2      | Culture awareness and understanding                        | SLR            |
| CSFP-3      | Personal relation building                                 | SLR            |
| CSFP-4      | Frequent interaction and negotiations                      | SLR            |
| CSFP-5      | Activity sharing about requirements                        | SLR            |
| CSFP-6      | Effective management practices and role of project manager | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B7** Practices for implementing “Culture Awareness.”

| Practice no | Practice                                                                               | Identified via |
|-------------|----------------------------------------------------------------------------------------|----------------|
| CSFP-1      | Training sessions and local workshops                                                  | SLR            |
| CSFP-2      | Computer-mediated communication for decreasing culture difference                      | SLR            |
| CSFP-3      | Need informal communication path                                                       | SLR            |
| CSFP-4      | Collaboration and coordination are needed                                              | SLR            |
| CSFP-5      | Regular reporting on culture awareness                                                 | SLR            |
| CSFP-6      | Translation activities are needed                                                      | SLR            |
| CSFP-7      | Establish cultural liaisons. To bridge the cultural gap across sites and organizations | SLR            |

Abbreviation: SLR, systematic literature review.

**TABLE B8** Practices for implementing “Coaching and Trainings.”

| Practice no | Practices                                                        | Identified via |
|-------------|------------------------------------------------------------------|----------------|
| CSFP-1      | English language skills are must                                 | SLR            |
| CSFP-2      | Computer mediated communication skills                           | SLR            |
| CSFP-3      | Training in “soft skills”                                        | SLR            |
| CSFP-4      | Professional trainings from certified organizations              | Questionnaire  |
| CSFP-5      | Train team members to use right process, tools, and technologies | Questionnaire  |

Abbreviation: SLR, systematic literature review.

**TABLE B9** Practices for implementing “Using collaborative tools and innovative technologies.”

| Practice no | Practice                                                                                                                    | Identified via |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|----------------|
| CSFP-1      | Select appropriate technology, according to both the environmental features and the stakeholders' cognitive characteristics | SLR            |
| CSFP-2      | Assure technology and tools access to all team members                                                                      | SLR            |
| CSFP-3      | Role of effective management and support                                                                                    | SLR            |
| CSFP-4      | Selection of modern tools from research                                                                                     | Questionnaire  |
| CSFP-5      | Selection of market oriented collaborative tools                                                                            | Questionnaire  |

Abbreviation: SLR, systematic literature review.

**TABLE B10** Practices for implementing “RE Process maturity.”

| Practice no | Practices                                                                         | Identified via |
|-------------|-----------------------------------------------------------------------------------|----------------|
| CSFP-1      | Use of agile process model                                                        | SLR            |
| CSFP-2      | Management support for introducing new RE process                                 | SLR            |
| CSFP-3      | Collaboration and feedback from other organizations on selecting RE process model | Questionnaire  |
| CSFP-4      | Adaptation of latest RE process activities from research                          | Questionnaire  |
| CSFP-5      | Apply best practices from the CMMI                                                | Questionnaire  |

Abbreviation: SLR, systematic literature review.

**TABLE B11** Practices for implementing “Agents and communication coordinator.”

| Practice no | Practice                                                           | Identified via |
|-------------|--------------------------------------------------------------------|----------------|
| CSFP-1      | Time management skills are required                                | Questionnaire  |
| CSFP-2      | Earn bachelor's degree in computer science or software engineering | Questionnaire  |
| CSFP-3      | Professionally certified in software engineering domain            | Questionnaire  |
| CSFP-4      | Ability to work both independently and as part of a team           | Questionnaire  |
| CSFP-5      | Excellent communication skills in English: orally and written.     | Questionnaire  |

**TABLE B12** Practices for implementing “Organizational infrastructure Improvement practice.”

| Practice no | Practices                                                                                        | Identified via |
|-------------|--------------------------------------------------------------------------------------------------|----------------|
| CSFP-1      | Research center should be established inside organization                                        | Questionnaire  |
| CSFP-2      | New business models should be introduced                                                         | Questionnaire  |
| CSFP-3      | Discover/uncover priority needs and requirements for organization                                | Questionnaire  |
| CSFP-4      | Create service area in organization                                                              | Questionnaire  |
| CSFP-5      | Set up an exploratory task force                                                                 | Questionnaire  |
| CSFP-6      | Skill development programs should be offered based on different expertise                        | Questionnaire  |
| CSFP-7      | Reduce documentation process as much possible in organizations                                   | Questionnaire  |
| CSFP-8      | Modularization of organizational infrastructure                                                  | Questionnaire  |
| CSFP-9      | Improve employee relationships with managers                                                     | Questionnaire  |
| CSFP-10     | Develop a better alternative to “Matrix Organization”                                            | Questionnaire  |
| CSFP-11     | Better match people to positions (talent management system)                                      | Questionnaire  |
| CSFP-12     | Develop a stronger organization planning and review system                                       | Questionnaire  |
| CSFP-13     | Improve organizational and project process management                                            | Questionnaire  |
| CSFP-14     | Personal software process and team software process models should be adopted from research       | Questionnaire  |
| CSFP-15     | Representative of different countries or continents should be kept in organizations              | Questionnaire  |
| CSFP-16     | Data analysis or analytic tools should be implemented for organizational performance measurement | Questionnaire  |
| CSFP-17     | Apply assessment tools such as CMMI as suggested by researchers                                  | Questionnaire  |

Abbreviation: CMMI, Capability Maturity Model Integration.



**TABLE B13** Practices for implementing “knowledge sharing and management.”

| Practice no | Practices                                                                                 | Identified via |
|-------------|-------------------------------------------------------------------------------------------|----------------|
| CSFP-1      | Effective communication channels among team members                                       | SLR            |
| CSFP-2      | Workshops and seminars on regular basis                                                   | SLR            |
| CSFP-3      | Experienced team members should be accessible                                             | SLR            |
| CSFP-4      | Who to talk to? Finding the right people                                                  | SLR            |
| CSFP-5      | Private email system is needed                                                            | SLR            |
| CSFP-6      | Common language to be used with-in the organization                                       | SLR            |
| CSFP-7      | Centralized communication structure can help new teams to remain aware                    | SLR            |
| CSFP-8      | Boost collaboration and reduce competition                                                | SLR            |
| CSFP-9      | Increase productivity for knowledge sharing                                               | SLR            |
| CSFP-10     | Get employees more flexible with their time                                               | Questionnaire  |
| CSFP-11     | Encouragement and foster knowledge sharing culture by the management (management support) | Questionnaire  |
| CSFP-12     | Regular meetings of senior experts with juniors                                           | Questionnaire  |
| CSFP-13     | Formalize a knowledge management process                                                  | Questionnaire  |

Abbreviation: SLR, systematic literature review.

**TABLE B14** Practices for implementing “Effective global project management.”

| Practice no | Practices                                                                             | Identified via |
|-------------|---------------------------------------------------------------------------------------|----------------|
| CSFP-1      | Global project management tools should be used                                        | SLR            |
| CSFP-2      | Clear definition of roles and responsibilities and make team wisely                   | SLR            |
| CSFP-3      | Apply best practices from the CMMI                                                    | SLR            |
| CSFP-4      | Information about best elicitation techniques used in market                          | SLR            |
| CSFP-5      | Obtaining information about the organizational environment                            | SLR            |
| CSFP-6      | Regular personal meetings with team members                                           | SLR            |
| CSFP-7      | Work with multiple partners. Distribute critical knowledge                            | SLR            |
| CSFP-8      | Have periodic workshops with teams.                                                   | SLR            |
| CSFP-9      | Formalization of documentation                                                        | SLR            |
| CSFP-10     | Develop a trust relationship in the team                                              | SLR            |
| CSFP-11     | Peer-to-peer links at all management                                                  | SLR            |
| CSFP-12     | Use a human facilitator in integrated rich communication media during decision making | SLR            |
| CSFP-13     | Maintain an organization risk repository                                              | SLR            |
| CSFP-14     | Organizations need to allow enough time to make the change                            | SLR            |
| CSFP-15     | Rigorously highlight insufficient quality                                             | SLR            |
| CSFP-16     | Get professional trainings on GSD                                                     | Questionnaire  |
| CSFP-17     | Preparation of requirements related challenges reports on regular basis               | Questionnaire  |
| CSFP-18     | Enhancing global project management using simulation                                  | Questionnaire  |

Abbreviations: GSD, global software development; SLR, systematic literature review.